

MAT 310, Handout 15: Gamma Distribution (4.6)

Gamma Distribution $X \sim \text{Gamma}(\alpha, \beta)$ is Gamma distributed with $\alpha > 0$ and $\beta > 0$ if its density function is given by

$$f(x) =$$

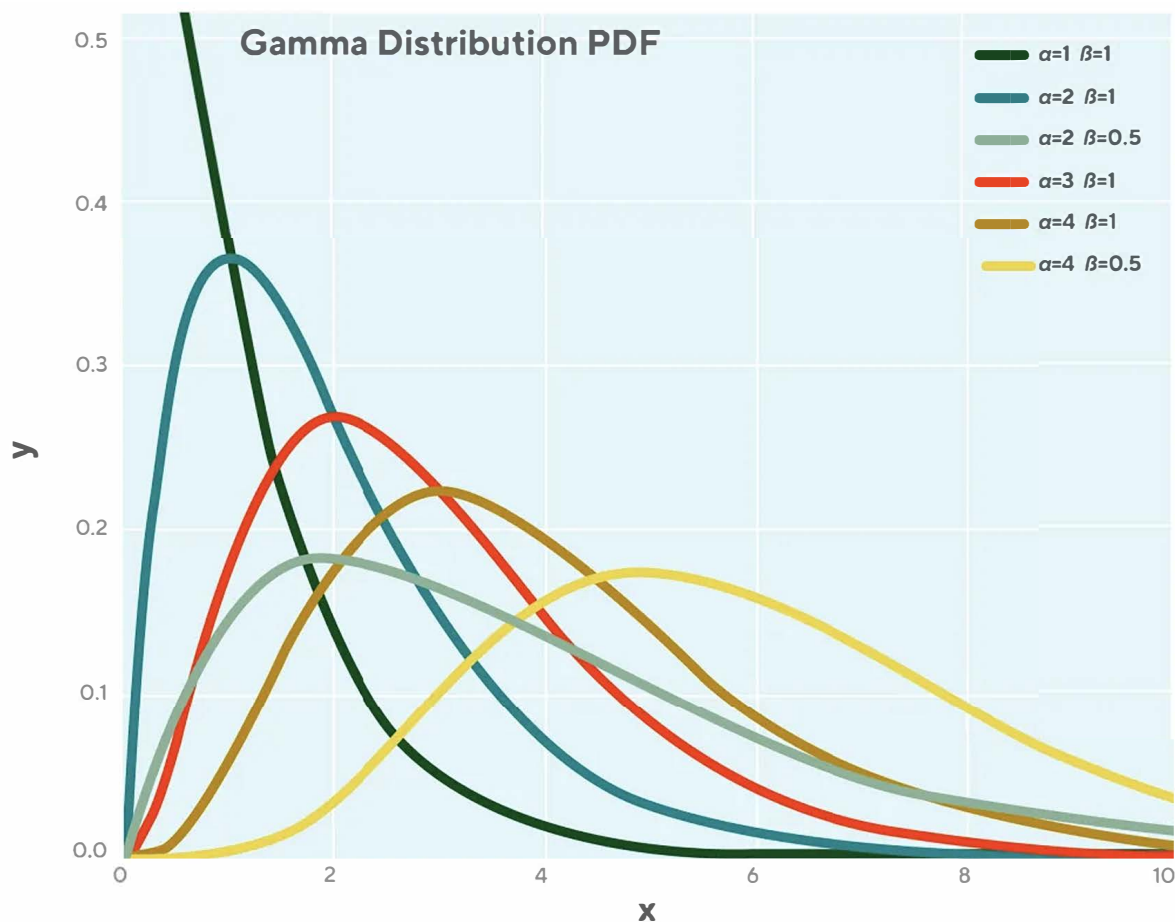
where the gamma function is given by $\Gamma(\alpha) =$

Remarks: The Gamma function has the following properties:

- $\Gamma(1) =$
- For $\alpha > 1$, $\Gamma(\alpha) =$
- For $\alpha \in \mathbb{Z}^+$, $\Gamma(\alpha) =$

- We refer to α as the _____ parameter and β as the _____ parameter.

Illustration (from DoorDash):



Expected Value and Variance of $X \sim \text{Gamma}(\alpha, \beta)$

$$E(X) =$$

$$V(X) =$$

Two Common Types of Gamma Distributions:

1. Chi-Square Distribution: $X \sim \chi^2$ with ν degrees of freedom (d.f.); i.e., $X \sim \chi_\nu^2$ or $X \sim \chi^2(\nu)$
 - Gamma RV with $\alpha = \nu/2, \beta = 2$
 - $E(X) =$
 - $V(X) =$
 - Useful in statistics (e.g., chi-square goodness of fit test is often used to evaluate whether sample data is representative of the population)
2. Exponential Distribution: $X \sim \exp(\beta)$
 - Gamma RV with $\alpha = 1$

Exponential Distribution $X \sim \exp(\beta)$ has a density function given by

$$f(x) =$$

Example 1. Find the cdf of $X \sim \exp(\beta)$.

Expected Value and Variance of $X \sim \exp(\beta)$

$$E(X) =$$

$$V(X) =$$

Proof of E(X):

Memoryless Property of Exponential RVs: If $X \sim \exp(\beta)$ and $a, b > 0$, then

$$P(X > a + b | X > a) =$$

Proof:

Example 2. Suppose the lifespan of light bulbs satisfies the memoryless property. Therefore, regardless of their respective ages, any 2 light bulbs have an _____ probability of lasting another b years.

Example 3. Suppose $Y \sim \exp(10)$. Find:

(a) $P(Y < 3)$

(b) $P(Y > 5)$

(c) $P(Y > 14|Y > 9)$

(d) $P(Y < 28|Y > 23)$

(e) the median of Y

(f) the 90th percentile of Y

Example 4. Suppose the length of light bulbs is exponentially distributed with mean of 1000 hours. If you buy 4 light bulbs, find the probability that at least one lasts for at least 2000 hours.